類　　科：冷凍空調工程技師
科　　目：冷凍空調設計
考試時間：二小時　　座號：_____

※注意：(1)不必抄題，作答時請將試題題號及答案依照順序寫在試卷上，於本試題上作答者，不予計分。
(2)使用電子計算器計算之試題，需詳列解答過程。

一、在一理想的單級蒸汽壓縮循環中，操作之冷凝溫度爲30°C，蒸發溫度爲 0 °C，若系統之冷凍能力爲30KW，分別以R-22及R-134a爲工作流體，計算下列各項：（25分）
㈠R-22與R-134a在各個狀態點下之熱力學性質（包括溫度、壓力、比容、比焓和比熵）。
㈡求出各冷媒之性能係數(COP)與質量流率。
㈢流過壓縮機的體積流率。
㈣設大氣溫度爲20°C，求出膨脹閥和壓縮機的不可逆性(irreversibility)。
㈤試討論兩種冷媒在相同的工作溫度條件下，以上計算之結果。

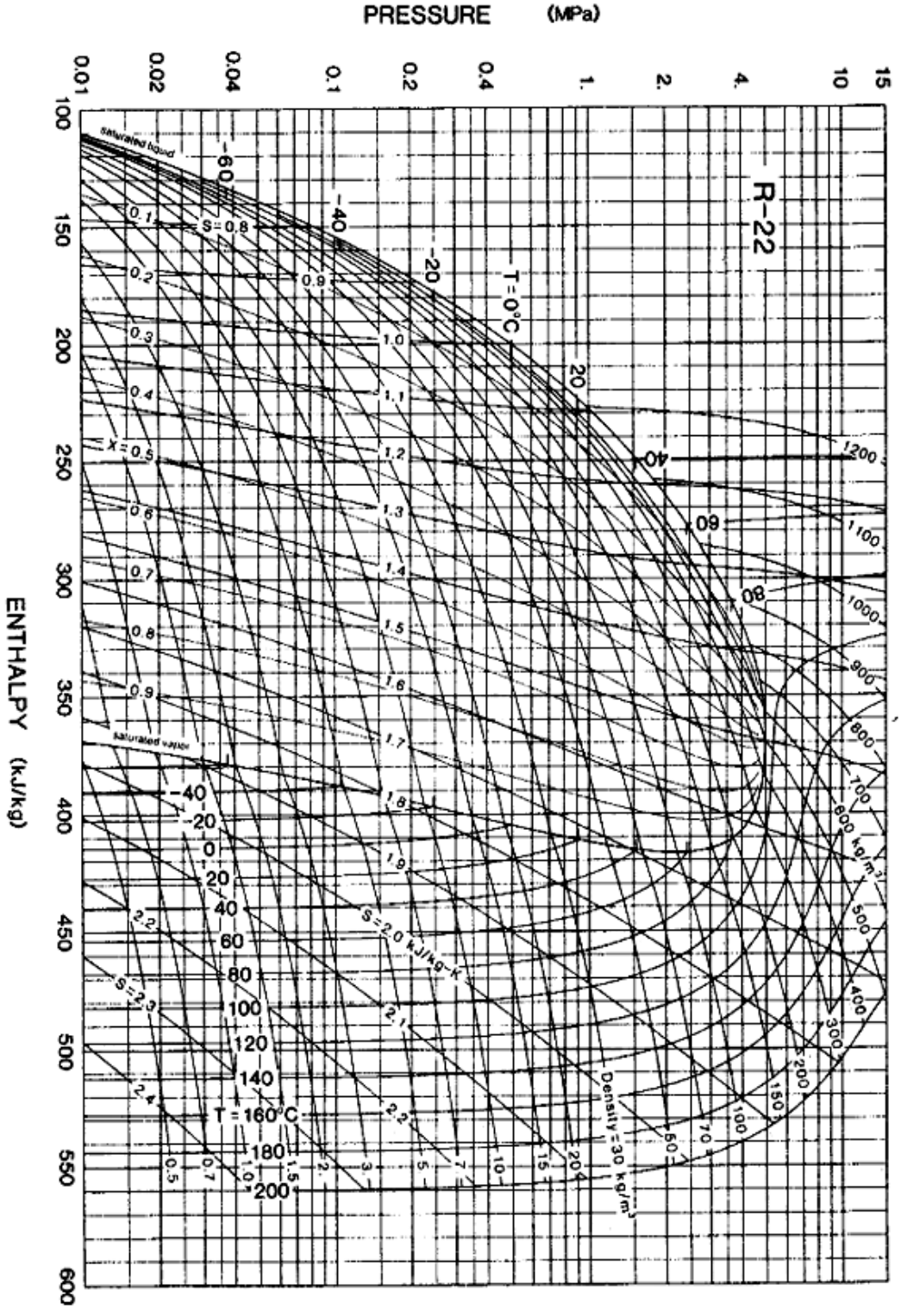
二、有一30KW，R-22螺旋式冰水機組，操作之冷凝與蒸發溫度分別爲30°C和0°C，今欲將工作流體R-22以R-134a取代，試回答下列問題：（25分）
㈠若螺旋式壓縮機不變（其設定容積比爲2.0），試問更換冷媒後壓縮機之體積流率和質量流率會如何改變？
㈡上題中，冷凍效果(refrigeration effect)和冷凍能力(refrigeration capacity)會如何改變？
㈢若R-134a和R-22之冷凝熱傳係數和蒸發熱傳係數十分接近，則冷凝器和蒸發器該如何修改？

三、㈠何謂風扇定律(fan law)？（5分）
㈡有一離心式風扇在空氣標準狀況下（101.325KPa，20°C，空氣密度1.20kg/m^3），風量爲3m^3/s，全壓差爲228KPa，轉速爲650rpm，耗電量爲0.855KW，試求：（20分）
(1)風扇之全壓效率。
(2)若風扇位於高山上操作（空氣密度爲1.0kg/m^3），則風量、全壓差、耗電量和全壓效率分別爲何？
(3)若風扇轉速減少爲600rpm，則風量、全壓差、耗電量和全壓效率又分別爲何？

四、一原先輸送冰水至室內側之水管路，因改裝儲冰式空調而改用乙烯乙二醇水溶液，於釋冷循環時，將水溶液送至負荷側，試問：（25分）
㈠若系統之管路、泵、閥等設備均不做修改，則室內側之冷卻效果會如何變化？
㈡若欲維持室內側達相同之冷卻效果，試舉出對系統管路、配件系統之修改方法。

（接下頁）

類　　科：冷凍空調工程技師
科　　目：冷凍空調設計



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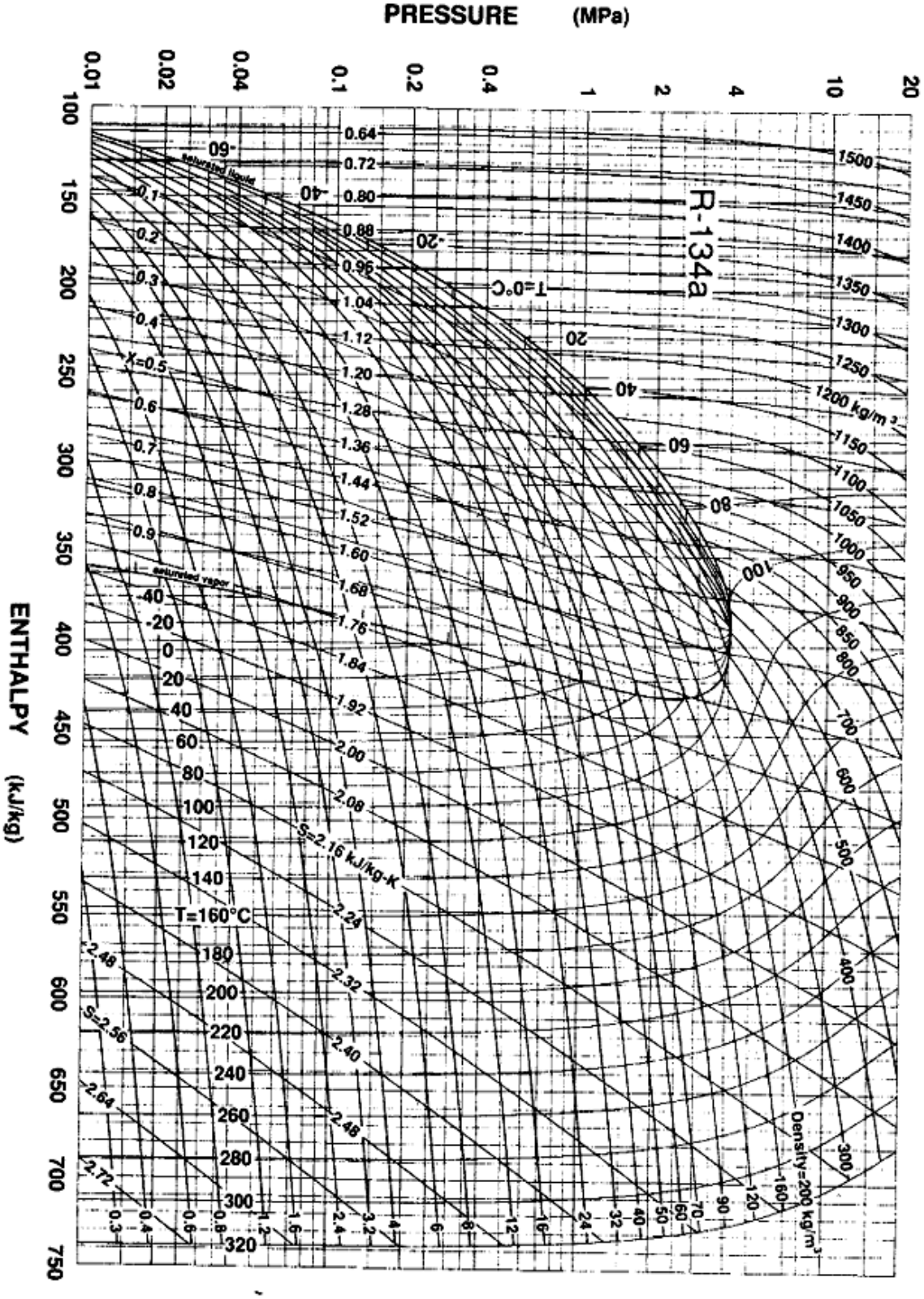
Refrigerant 22 (Chlorodifluoromethane) Properties of Saturated Liquid and Saturated Vapor

Temp.,* °C	Absolute Pressure, MPa	Density, kg/m³ Liquid	Volume, m³/kg Vapor	Enthalpy, kJ/kg Liquid	Enthalpy, kJ/kg Vapor	Entropy, kJ/(kg·K) Liquid	Entropy, kJ/(kg·K) Vapor	Specific Heat c_p, kJ/(kg·K) Liquid	Specific Heat c_p, kJ/(kg·K) Vapor	c_p/c_v, Vapor	Velocity of Sound, m/s Liquid	Velocity of Sound, m/s Vapor	Viscosity, μPa·s Liquid	Viscosity, μPa·s Vapor	Thermal Cond, mW/(m·K) Liquid	Thermal Cond, mW/(m·K) Vapor	Surface Tension, mN/m	Temp.,* °C
−150.00	–	1701.5	–	26.01	335.85	0.0566	2.5752	–	0.434	1.285	–	125.	–	–	–	–	37.59	−150.00
−140.00	–	1675.3	–	43.84	340.24	0.1961	2.4222	–	0.445	1.275	–	128.	–	–	–	–	35.70	−140.00
−130.00	0.00006	1649.7	239.29	57.00	344.75	0.2916	2.3017	–	0.458	1.266	–	132.	–	–	–	–	33.84	−130.00
−120.00	0.00023	1624.0	65.648	68.51	349.38	0.3694	2.2033	–	0.470	1.258	–	136.	–	–	–	–	32.00	−120.00
−110.00	0.00074	1598.0	21.311	79.47	354.11	0.4386	2.1220	–	0.485	1.250	–	140.	–	–	–	–	30.17	−110.00
−100.00	0.00200	1571.7	8.2980	90.24	358.93	0.5027	2.0545	–	0.497	1.243	–	144.	–	–	–	–	28.37	−100.00
−90.00	0.00480	1545.1	3.6548	100.95	363.82	0.5629	1.9982	1.070	0.511	1.237	1094.	147.	–	–	–	–	26.59	−90.00
−80.00	0.01035	1518.3	1.7816	111.66	368.75	0.6197	1.9508	1.070	0.527	1.233	1037.	150.	–	–	–	–	24.83	−80.00
−70.00	0.02044	1491.1	0.94476	122.36	373.68	0.6738	1.9109	1.072	0.544	1.231	986.	153.	–	–	128.0	–	23.10	−70.00
−60.00	0.03747	1463.6	0.53734	133.11	378.58	0.7253	1.8770	1.076	0.563	1.231	937.	156.	–	–	123.1	5.61	21.39	−60.00
−50.00	0.06449	1435.5	0.32405	143.91	383.39	0.7748	1.8480	1.083	0.584	1.233	890.	158.	–	–	118.4	6.31	19.70	−50.00
−48.00	0.07140	1429.8	0.29469	146.08	384.35	0.7844	1.8427	1.085	0.589	1.233	881.	159.	–	–	117.5	6.44	19.37	−48.00
−46.00	0.07890	1424.1	0.26849	148.25	385.29	0.7940	1.8376	1.087	0.594	1.234	871.	159.	–	–	116.5	6.58	19.04	−46.00
−44.00	0.08700	1418.4	0.24507	150.43	386.23	0.8035	1.8326	1.089	0.598	1.235	862.	160.	–	–	115.6	6.71	18.70	−44.00
−42.00	0.09575	1412.6	0.22410	152.61	387.17	0.8130	1.8277	1.091	0.603	1.236	853.	160.	–	–	114.7	6.85	18.37	−42.00
−40.80b	0.10132	1409.1	0.21256	153.93	387.72	0.8186	1.8249	1.092	0.606	1.237	847.	160.	–	–	114.1	6.93	18.18	−40.80
−40.00	0.10518	1406.8	0.20526	154.80	388.09	0.8224	1.8230	1.093	0.608	1.237	844.	160.	–	–	113.8	6.98	18.05	−40.00
−38.00	0.11533	1401.0	0.18852	156.99	389.01	0.8317	1.8184	1.096	0.614	1.239	834.	161.	–	–	112.9	7.11	17.72	−38.00
−36.00	0.12623	1395.1	0.17306	159.19	389.93	0.8410	1.8140	1.098	0.619	1.240	825.	161.	–	–	112.0	7.24	17.39	−36.00
−34.00	0.13793	1389.2	0.15927	161.40	390.84	0.8502	1.8096	1.101	0.624	1.242	816.	161.	–	–	111.1	7.37	17.07	−34.00
−32.00	0.15045	1383.3	0.14680	163.61	391.74	0.8594	1.8054	1.104	0.630	1.243	807.	161.	–	–	110.1	7.50	16.74	−32.00
−30.00	0.16384	1377.3	0.13551	165.82	392.63	0.8685	1.8013	1.107	0.636	1.245	797.	162.	–	–	109.2	7.63	16.42	−30.00
−28.00	0.17815	1371.3	0.12525	168.04	393.52	0.8776	1.7973	1.110	0.642	1.247	788.	162.	–	–	108.4	7.76	16.10	−28.00
−26.00	0.19340	1365.2	0.11593	170.27	394.39	0.8866	1.7934	1.114	0.648	1.249	779.	162.	–	–	107.5	7.89	15.77	−26.00
−24.00	0.20965	1359.1	0.10744	172.51	395.26	0.8955	1.7896	1.117	0.654	1.252	770.	162.	–	–	106.6	8.02	–	−24.00
−22.00	0.22693	1352.9	0.09970	174.75	396.12	0.9044	1.7859	1.121	0.660	1.254	760.	163.	–	–	105.7	8.14	–	−22.00
−20.00	0.24529	1346.8	0.09262	177.00	396.97	0.9133	1.7822	1.125	0.667	1.257	751.	163.	260.1	–	104.8	8.27	–	−20.00
−18.00	0.26477	1340.5	0.08615	179.26	397.81	0.9222	1.7787	1.129	0.674	1.260	742.	163.	254.7	11.08	103.9	8.40	–	−18.00
−16.00	0.28542	1334.2	0.08023	181.53	398.64	0.9309	1.7752	1.133	0.681	1.263	733.	163.	249.4	11.16	103.1	8.52	–	−16.00
−14.00	0.30728	1327.9	0.07479	183.81	399.46	0.9397	1.7719	1.137	0.688	1.266	725.	163.	244.2	11.24	102.2	8.64	–	−14.00
−12.00	0.33040	1321.5	0.06979	186.09	400.27	0.9484	1.7686	1.141	0.695	1.269	716.	163.	239.1	11.32	101.3	8.77	–	−12.00
−10.00	0.35482	1315.0	0.06520	188.38	401.07	0.9571	1.7653	1.146	0.703	1.273	705.	163.	234.1	11.40	100.4	8.89	–	−10.00
−8.00	0.38059	1308.5	0.06096	190.69	401.85	0.9657	1.7621	1.151	0.710	1.277	696.	163.	229.1	11.48	99.6	9.02	–	−8.00
−6.00	0.40775	1301.9	0.05706	193.00	402.63	0.9743	1.7590	1.156	0.718	1.281	686.	163.	224.2	11.56	98.7	9.14	–	−6.00
−4.00	0.43636	1295.3	0.05345	195.32	403.39	0.9829	1.7560	1.161	0.727	1.285	677.	163.	219.4	11.64	97.9	9.26	–	−4.00
−2.00	0.46646	1288.6	0.05012	197.66	404.14	0.9915	1.7530	1.166	0.735	1.289	668.	163.	214.7	11.72	97.0	9.38	–	−2.00
0.00	0.49811	1281.8	0.04703	200.00	404.87	1.0000	1.7500	1.171	0.744	1.294	658.	163.	210.1	11.80	96.2	9.50	–	0.00
2.00	0.53134	1275.0	0.04417	202.35	405.59	1.0085	1.7471	1.177	0.753	1.299	649.	163.	205.6	11.88	95.3	9.63	–	2.00
4.00	0.56622	1268.1	0.04152	204.72	406.30	1.0170	1.7443	1.183	0.762	1.305	640.	163.	201.2	11.96	94.5	9.75	–	4.00
6.00	0.60279	1261.1	0.03906	207.10	406.99	1.0254	1.7415	1.189	0.772	1.310	630.	163.	196.9	12.04	93.6	9.87	–	6.00
8.00	0.64109	1254.0	0.03676	209.49	407.67	1.0338	1.7387	1.195	0.782	1.316	621.	163.	192.6	12.12	92.8	9.99	–	8.00
10.00	0.68119	1246.9	0.03463	211.89	408.33	1.0422	1.7360	1.202	0.792	1.323	611.	163.	188.5	12.20	92.0	10.11	–	10.00
12.00	0.72314	1239.7	0.03265	214.31	408.97	1.0506	1.7333	1.208	0.802	1.330	602.	162.	184.4	12.28	91.1	10.23	–	12.00
14.00	0.76698	1232.4	0.03079	216.74	409.60	1.0590	1.7306	1.215	0.813	1.337	592.	162.	180.5	12.36	90.3	10.35	–	14.00
16.00	0.81277	1225.0	0.02906	219.18	410.21	1.0675	1.7280	1.223	0.825	1.345	583.	162.	176.6	12.44	89.5	10.47	–	16.00
18.00	0.86056	1217.6	0.02744	221.63	410.80	1.0756	1.7254	1.230	0.837	1.353	573.	162.	172.8	12.52	88.7	10.59	–	18.00
20.00	0.91041	1210.0	0.02593	224.10	411.38	1.0840	1.7228	1.238	0.849	1.361	564.	161.	169.1	–	87.8	10.71	–	20.00
22.00	0.96236	1202.4	0.02451	226.59	411.93	1.0923	1.7202	1.246	0.862	1.370	554.	161.	165.4	–	87.0	10.82	–	22.00
24.00	1.0165	1194.6	0.02319	229.09	412.46	1.1006	1.7177	1.254	0.875	1.380	544.	160.	161.9	–	86.2	10.94	–	24.00
26.00	1.0728	1186.8	0.02194	231.60	412.98	1.1088	1.7151	1.263	0.889	1.391	535.	160.	158.4	–	85.4	11.06	–	26.00
28.00	1.1314	1178.8	0.02077	234.14	413.46	1.1171	1.7126	1.272	0.904	1.402	525.	160.	155.0	–	84.6	11.18	–	28.00
30.00	1.1924	1170.7	0.01968	236.69	413.95	1.1254	1.7101	1.282	0.919	1.413	515.	159.	151.7	–	83.8	11.30	–	30.00
32.00	1.2557	1162.5	0.01864	239.25	414.37	1.1336	1.7075	1.292	0.935	1.426	506.	159.	148.5	–	83.0	11.42	–	32.00
34.00	1.3215	1154.2	0.01767	241.84	414.79	1.1419	1.7050	1.302	0.952	1.440	496.	158.	145.4	–	82.2	11.54	–	34.00
36.00	1.3898	1145.7	0.01675	244.44	415.18	1.1501	1.7024	1.313	0.970	1.454	486.	158.	142.3	–	81.4	11.66	–	36.00
38.00	1.4606	1137.1	0.01589	247.06	415.54	1.1584	1.6999	1.325	0.989	1.470	476.	157.	139.3	–	80.6	11.78	–	38.00
40.00	1.5341	1128.4	0.01507	249.71	415.87	1.1667	1.6975	1.338	1.009	1.486	466.	156.	136.3	–	79.8	11.90	–	40.00
42.00	1.6103	1119.5	0.01430	252.37	416.17	1.1749	1.6947	1.351	1.030	1.504	456.	156.	–	–	79.0	12.02	–	42.00
44.00	1.6892	1110.4	0.01357	255.06	416.44	1.1832	1.6921	1.365	1.052	1.524	446.	155.	–	–	78.2	12.14	–	44.00
46.00	1.7709	1101.2	0.01288	257.77	416.68	1.1915	1.6894	1.380	1.076	1.545	436.	154.	–	–	77.4	12.26	–	46.00
48.00	1.8555	1091.8	0.01223	260.51	416.87	1.1998	1.6867	1.396	1.102	1.568	426.	153.	–	–	76.6	12.38	–	48.00
50.00	1.9431	1082.1	0.01161	263.27	417.04	1.2081	1.6840	1.414	1.129	1.593	415.	153.	–	–	–	–	–	50.00
55.00	2.1753	1057.1	0.01020	270.31	417.24	1.2291	1.6768	1.464	1.209	1.667	389.	150.	–	–	–	–	–	55.00
60.00	2.4274	1030.5	0.00895	277.56	417.14	1.2503	1.6692	1.528	1.307	1.762	362.	148.	–	–	–	–	–	60.00
65.00	2.7008	1001.8	0.00784	285.06	416.65	1.2718	1.6610	1.611	1.435	1.888	335.	145.	–	–	–	–	–	65.00
70.00	2.9967	970.4	0.00684	292.90	415.69	1.2940	1.6518	1.726	1.609	2.064	306.	142.	–	–	–	–	–	70.00
75.00	3.3168	935.3	0.00594	301.18	414.09	1.3169	1.6413	1.896	1.862	2.324	276.	138.	–	–	–	–	–	75.00
80.00	3.6627	894.8	0.00511	310.10	411.60	1.3413	1.6287	2.176	2.268	2.748	244.	134.	–	–	–	–	–	80.00
85.00	4.0368	845.1	0.00433	320.05	407.72	1.3680	1.6128	2.725	3.049	5.555	209.	129.	–	–	–	–	–	85.00
90.00	4.4416	777.5	0.00355	331.98	401.33	1.3998	1.5907	4.186	5.041	5.643	170.	124.	–	–	–	–	–	90.00
95.00	4.8820	665.4	0.00264	348.86	387.46	1.4442	1.5491	∞	∞	∞	0.	0.	–	–	∞	∞	0.00	95.00
96.14c	4.9900	523.8	0.00191	366.59	366.59	1.4918	1.4918	∞	∞	∞	0.	0.	–	–	∞	∞	0.00	96.14

*Temperatures are on the ITS–90 scale　　b = normal boiling point　　c = critical point

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類　　科：冷凍空調工程技師
科　　目：冷凍空調設計



類　　科：冷凍空調工程技師
科　　目：冷凍空調設計

Refrigerant 134a (1,1,1,2-Tetrafluoroethane) Properties of Saturated Liquid and Saturated Vapor

Temp.,[a] °C	Absolute Pressure, MPa	Density, kg/m³ Liquid	Volume, m³/kg Vapor	Enthalpy, kJ/kg Liquid	Enthalpy, kJ/kg Vapor	Entropy, kJ/(kg·K) Liquid	Entropy, kJ/(kg·K) Vapor	Specific Heat c_p, kJ/(kg·K) Liquid	Specific Heat c_p, kJ/(kg·K) Vapor	c_p/c_v Vapor	Velocity of Sound, m/s Liquid	Velocity of Sound, m/s Vapor	Viscosity, μPa·s Liquid	Viscosity, μPa·s Vapor	Thermal Cond, mW/(m·K) Liquid	Thermal Cond, mW/(m·K) Vapor	Surface Tension, mN/m	Temp.,[a] °C
−103.30a	0.00039	1591.2	35.263	71.89	335.07	0.4143	1.9658	1.147	0.585	1.163	1135.	127.	2186.6	6.65	–	–	28.15	−103.30
−100.00	0.00056	1581.9	25.039	75.71	337.00	0.4366	1.9456	1.168	0.592	1.161	1111.	128.	1958.2	6.76	–	–	27.56	−100.00
−90.00	0.00153	1553.9	9.7191	87.59	342.94	0.5032	1.8975	1.201	0.614	1.155	1051.	131.	1445.6	7.16	–	–	25.81	−90.00
−80.00	0.00369	1526.2	4.2504	99.65	349.03	0.5674	1.8585	1.211	0.637	1.151	999.	134.	1109.9	7.57	–	–	24.11	−80.00
−70.00	0.00801	1498.6	2.0528	111.78	355.23	0.6286	1.8269	1.215	0.660	1.148	951.	137.	879.6	7.97	125.8	–	22.44	−70.00
−60.00	0.01594	1471.0	1.0770	123.96	361.51	0.6871	1.8016	1.220	0.685	1.146	904.	139.	715.4	8.38	121.1	–	20.81	−60.00
−50.00	0.02948	1443.1	0.60560	136.21	367.85	0.7432	1.7812	1.229	0.712	1.146	858.	142.	594.3	8.79	116.5	7.12	19.22	−50.00
−40.00	0.05122	1414.8	0.36095	148.57	374.16	0.7973	1.7649	1.243	0.740	1.148	812.	144.	502.2	9.20	111.9	8.19	17.66	−40.00
−30.00	0.08436	1385.9	0.22596	161.10	380.45	0.8498	1.7519	1.260	0.771	1.152	765.	145.	430.4	9.62	107.3	9.16	16.13	−30.00
−28.00	0.09268	1380.0	0.20682	163.62	381.70	0.8601	1.7497	1.264	0.778	1.155	756.	145.	418.0	9.71	106.3	9.35	15.83	−28.00
−26.07b	0.10132	1374.3	0.19016	166.07	382.90	0.8701	1.7476	1.268	0.784	1.154	747.	146.	406.4	9.79	105.4	9.52	15.54	−26.07
−26.00	0.10164	1374.1	0.18961	166.16	382.94	0.8704	1.7476	1.268	0.785	1.154	747.	146.	406.0	9.79	105.4	9.53	15.53	−26.00
−24.00	0.11127	1368.2	0.17410	168.70	384.19	0.8806	1.7455	1.273	0.791	1.155	738.	146.	394.6	9.88	104.5	9.71	15.23	−24.00
−22.00	0.12160	1362.2	0.16010	171.26	385.43	0.8908	1.7436	1.277	0.798	1.156	728.	146.	383.6	9.96	103.6	9.89	14.93	−22.00
−20.00	0.13268	1356.2	0.14744	173.82	386.66	0.9009	1.7417	1.282	0.805	1.157	719.	146.	373.1	10.05	102.6	10.07	14.63	−20.00
−18.00	0.14454	1350.2	0.13597	176.39	387.89	0.9110	1.7399	1.286	0.812	1.159	710.	146.	363.0	10.14	101.7	10.24	14.33	−18.00
−16.00	0.15721	1344.1	0.12556	178.97	389.11	0.9211	1.7383	1.291	0.820	1.160	700.	147.	353.3	10.22	100.8	10.42	14.04	−16.00
−14.00	0.17074	1338.0	0.11610	181.56	390.33	0.9311	1.7367	1.296	0.827	1.162	691.	147.	344.0	10.31	99.9	10.59	13.74	−14.00
−12.00	0.18516	1331.8	0.10749	184.16	391.55	0.9410	1.7351	1.301	0.835	1.164	682.	147.	335.0	10.40	99.0	10.76	13.45	−12.00
−10.00	0.20052	1325.6	0.09963	186.78	392.75	0.9509	1.7337	1.306	0.842	1.166	672.	147.	326.3	10.49	98.0	10.93	13.16	−10.00
−8.00	0.21684	1319.3	0.09246	189.40	393.95	0.9608	1.7323	1.312	0.850	1.168	663.	147.	318.0	10.58	97.1	11.10	12.87	−8.00
−6.00	0.23418	1313.0	0.08591	192.03	395.15	0.9707	1.7310	1.317	0.858	1.170	654.	147.	309.9	10.67	96.2	11.28	12.58	−6.00
−4.00	0.25257	1306.6	0.07991	194.68	396.33	0.9805	1.7297	1.323	0.866	1.172	644.	147.	302.2	10.76	95.3	11.45	12.29	−4.00
−2.00	0.27206	1300.2	0.07440	197.33	397.51	0.9903	1.7285	1.329	0.875	1.175	635.	147.	294.7	10.85	94.3	11.62	12.00	−2.00
0.00	0.29269	1293.7	0.06955	200.00	398.68	1.0000	1.7274	1.335	0.883	1.178	626.	147.	287.4	10.94	93.4	11.79	11.71	0.00
2.00	0.31450	1287.1	0.06470	202.68	399.84	1.0097	1.7263	1.341	0.892	1.180	616.	147.	280.4	11.03	92.5	11.96	11.43	2.00
4.00	0.33755	1280.5	0.06042	205.37	401.00	1.0194	1.7252	1.347	0.901	1.183	607.	147.	273.6	11.13	91.6	12.13	11.14	4.00
6.00	0.36186	1275.8	0.05648	208.08	402.14	1.0291	1.7242	1.353	0.910	1.187	598.	147.	267.0	11.22	90.7	12.31	10.86	6.00
8.00	0.38749	1267.0	0.05284	210.80	403.27	1.0387	1.7233	1.360	0.920	1.190	588.	147.	260.6	11.32	89.7	12.48	10.58	8.00
10.00	0.41449	1260.2	0.04948	213.53	404.40	1.0483	1.7224	1.367	0.930	1.193	579.	146.	254.3	11.42	88.8	12.66	10.30	10.00
12.00	0.44289	1253.3	0.04636	216.27	405.51	1.0579	1.7215	1.374	0.939	1.197	569.	146.	248.3	11.52	87.9	12.84	10.02	12.00
14.00	0.47276	1246.3	0.04348	219.03	406.61	1.0674	1.7207	1.381	0.950	1.201	560.	146.	242.5	11.62	87.0	13.02	9.74	14.00
16.00	0.50413	1239.3	0.04081	221.80	407.70	1.0770	1.7199	1.388	0.960	1.206	550.	146.	236.8	11.72	86.0	13.20	9.47	16.00
18.00	0.53706	1232.1	0.03853	224.59	408.78	1.0865	1.7191	1.396	0.971	1.210	541.	146.	231.2	11.82	85.1	13.39	9.19	18.00
20.00	0.57159	1224.9	0.03603	227.40	409.84	1.0960	1.7183	1.404	0.982	1.215	532.	145.	225.8	11.92	84.2	13.57	8.92	20.00
22.00	0.60777	1217.5	0.03388	230.21	410.89	1.1055	1.7176	1.412	0.994	1.220	522.	145.	220.5	12.03	83.3	13.76	8.65	22.00
24.00	0.64566	1210.1	0.03189	233.05	411.93	1.1149	1.7169	1.420	1.006	1.226	512.	145.	215.4	12.14	82.4	13.96	8.38	24.00
26.00	0.68531	1202.6	0.03003	235.90	412.95	1.1244	1.7162	1.429	1.018	1.231	503.	144.	210.4	12.25	81.4	14.15	8.11	26.00
28.00	0.72676	1194.9	0.02829	238.77	413.95	1.1338	1.7155	1.438	1.031	1.238	493.	144.	205.5	12.36	80.5	14.35	7.84	28.00
30.00	0.77008	1187.2	0.02667	241.65	414.94	1.1432	1.7149	1.447	1.044	1.244	484.	143.	200.7	12.48	79.6	14.56	7.57	30.00
32.00	0.81530	1179.3	0.02516	244.55	415.90	1.1527	1.7142	1.457	1.058	1.251	474.	143.	196.0	12.60	78.7	14.76	7.31	32.00
34.00	0.86250	1171.3	0.02374	247.47	416.85	1.1621	1.7135	1.467	1.075	1.259	465.	142.	191.4	12.72	77.7	14.97	7.05	34.00
36.00	0.91172	1163.2	0.02241	250.41	417.78	1.1715	1.7129	1.478	1.088	1.267	455.	142.	186.9	12.84	76.8	15.19	6.78	36.00
38.00	0.96301	1154.9	0.02116	253.37	418.69	1.1809	1.7122	1.489	1.104	1.276	445.	141.	182.5	12.97	75.9	15.41	6.52	38.00
40.00	1.0165	1146.5	0.01999	256.35	419.58	1.1903	1.7115	1.500	1.120	1.285	436.	140.	178.2	13.10	75.0	15.64	6.27	40.00
42.00	1.0721	1137.9	0.01890	259.35	420.44	1.1997	1.7108	1.513	1.138	1.295	426.	140.	174.0	13.24	74.1	15.86	6.01	42.00
44.00	1.1300	1129.2	0.01786	262.38	421.28	1.2091	1.7101	1.525	1.156	1.306	416.	139.	169.8	13.38	73.1	16.10	5.76	44.00
46.00	1.1901	1120.3	0.01689	265.42	422.09	1.2185	1.7094	1.539	1.175	1.318	407.	138.	165.7	13.52	72.2	16.34	5.51	46.00
48.00	1.2527	1111.3	0.01598	268.49	422.88	1.2279	1.7086	1.553	1.196	1.331	397.	137.	161.7	13.67	71.3	16.59	5.26	48.00
50.00	1.3177	1102.0	0.01511	271.59	423.65	1.2373	1.7078	1.569	1.218	1.345	387.	137.	157.7	13.85	70.4	16.84	5.01	50.00
52.00	1.3852	1092.6	0.01430	274.71	424.35	1.2468	1.7070	1.585	1.241	1.360	377.	136.	153.8	13.99	69.5	17.10	4.76	52.00
54.00	1.4553	1082.9	0.01353	277.86	425.03	1.2562	1.7061	1.602	1.266	1.377	367.	135.	149.9	14.16	68.5	17.36	4.52	54.00
56.00	1.5280	1073.0	0.01280	281.04	425.68	1.2657	1.7051	1.621	1.293	1.395	358.	134.	146.1	14.35	67.6	17.63	4.28	56.00
58.00	1.6033	1062.8	0.01212	284.25	426.29	1.2752	1.7041	1.641	1.322	1.416	348.	133.	142.3	14.51	66.7	17.91	4.04	58.00
60.00	1.6815	1052.4	0.01146	287.49	426.86	1.2847	1.7031	1.663	1.354	1.438	338.	132.	138.6	14.71	65.8	18.19	3.81	60.00
62.00	1.7625	1041.7	0.01085	290.77	427.37	1.2943	1.7019	1.686	1.388	1.463	328.	131.	134.9	14.91	64.9	18.48	3.57	62.00
64.00	1.8464	1030.7	0.01026	294.08	427.84	1.3039	1.7007	1.712	1.426	1.490	318.	129.	131.2	15.12	63.9	18.78	3.34	64.00
66.00	1.9334	1019.4	0.00970	297.44	428.25	1.3136	1.6993	1.740	1.468	1.522	308.	128.	127.5	15.35	63.0	19.09	3.12	66.00
68.00	2.0234	1007.7	0.00917	300.84	428.61	1.3234	1.6979	1.772	1.515	1.557	298.	127.	123.9	15.59	62.1	19.40	2.89	68.00
70.00	2.1165	995.6	0.00867	304.29	428.89	1.3332	1.6963	1.806	1.567	1.597	287.	126.	120.3	15.85	61.2	19.72	2.67	70.00
72.00	2.2130	983.1	0.00818	307.79	429.10	1.3430	1.6945	1.846	1.626	1.642	277.	124.	116.7	16.12	60.3	20.05	2.46	72.00
74.00	2.3127	970.0	0.00772	311.34	429.23	1.3530	1.6926	1.890	1.693	1.695	267.	123.	113.1	16.41	59.3	20.39	2.24	74.00
76.00	2.4159	956.5	0.00728	314.96	429.27	1.3631	1.6905	1.941	1.770	1.757	256.	121.	109.4	16.73	58.4	20.74	2.03	76.00
78.00	2.5227	942.3	0.00686	318.65	429.20	1.3733	1.6881	2.000	1.861	1.830	246.	120.	105.8	17.08	57.5	21.09	1.85	78.00
80.00	2.6331	927.4	0.00646	322.41	429.02	1.3837	1.6855	2.069	1.967	1.917	235.	118.	102.1	17.46	56.6	21.46	1.63	80.00
85.00	2.9259	886.2	0.00550	332.27	427.91	1.4105	1.6775	2.315	2.348	2.231	207.	113.	92.7	18.59	54.5	22.41	1.15	85.00
90.00	3.2445	836.9	0.00461	343.01	425.48	1.4392	1.6663	2.766	3.064	2.832	178.	108.	82.6	20.15	–	–	0.72	90.00
95.00	3.5916	771.6	0.00374	355.43	420.60	1.4720	1.6490	5.961	4.942	4.424	145.	102.	70.9	22.59	–	–	0.35	95.00
100.00	3.9721	646.7	0.00265	374.02	407.08	1.5207	1.6093	–	–	–	105.	94.	53.0	28.86	–	–	0.03	100.00
101.03c	4.0560	513.3	0.00195	389.79	389.79	1.5593	1.5593	∞	∞	∞	0	0	–	–	∞	∞	0	101.05

[a]Temperatures are on the ITS-90 scale　　a = triple point　　b = normal boiling point　　c = critical point